Chapter 13. Indicator Variables

Dummy variables, interactions, several categories, piecewise linear regression

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13.1 One indicator variable

Qualitative explanatory variables (sex, treatment, region) enter a regression through **indicator** (dummy) variables taking the values 0 and 1.

Parallel lines. Suppose y is the score on a test, x_1 the hours of preparation, and $x_2 = 1$ for students who attended a review session and 0 otherwise. The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

gives two response functions with a common slope:

$$E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + \beta_1 x_1, & x_2 = 1. \end{cases}$$

β_1 is the effect of x_1 in both groups, and β_2 is the vertical distance between the parallel lines: the group effect at any fixed x_1. Fitting one model, rather than one line per group, uses all n observations to estimate the common slope and σ^2.

Interaction. If the effect of x_1 may differ between groups, add the product term:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i, \qquad E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x_1, & x_2 = 1. \end{cases}$$

Now β_2 is the difference in intercepts and β_3 the difference in slopes. The hypotheses of Chapter 5 become ordinary tests in one model:

- parallel lines: $H_0 : \beta_3 = 0$ (a t test);
- identical lines: $H_0 : \beta_2 = \beta_3 = 0$ (a partial F test with 2 degrees of freedom).

These agree with the separate-fit tests of Section 5.3 because the interaction model with $n_1 + n_2$ observations has exactly the four parameters of the two separate lines and the same residual sum of squares.

Example 13.1

In a study of 20 firms, y is the time to adopt an innovation, x_1 the firm size and $x_2 = 1$ for stock companies. The fit $\hat{y} = 33.87 - 0.102 x_1 + 8.06 x_2$ says that adoption is faster for larger firms (about 1 month per 10 size units) and that stock companies take about 8 months longer at every size. Adding the interaction gives $\hat{\beta}_3 = -0.0004$ with standard error 0.018, so $t_0 = -0.02$: the parallel-lines model is adequate.

13.2 Several categories

A qualitative variable with q categories needs $q - 1$ indicators. With three categories E_1, E_2, E_3, define

$$x_2 = \mathbf{1}\{E_1\}, \qquad x_3 = \mathbf{1}\{E_2\},$$

with E_3 the **baseline**:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i, \qquad E(y) = \begin{cases} (\beta_0 + \beta_2) + \beta_1 x_1, & E_1, \\ (\beta_0 + \beta_3) + \beta_1 x_1, & E_2, \\ \beta_0 + \beta_1 x_1, & E_3. \end{cases}$$

So β_2 and β_3 compare E_1 and E_2 with the baseline, and $\beta_2 - \beta_3$ compares E_1 with E_2.

The dummy variable trap. Using q indicators together with an intercept makes the columns of X linearly dependent: the indicators sum to the intercept column $\mathbf{1}$. Then $\operatorname{rank} X <$ number of columns, $X'X$ is singular, and β is not identifiable. Either drop one indicator (baseline coding, as above), drop the intercept, or impose a constraint such as $\sum(\text{effects}) = 0$.

Interactions and several factors. Separate slopes are obtained with the products x_1x_2 and x_1x_3:

$$E(y) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_1x_2 + \beta_5x_1x_3,$$

so that in E_1 the line is $(\beta_0+\beta_2)+(\beta_1+\beta_4)x_1$, and so on. Several qualitative factors each get their own set of indicators, and interactions between factors are products of their indicators. A model containing only indicators (no quantitative regressors) is an analysis of variance model written as a regression.

13.3 Piecewise linear regression

Sometimes the slope changes at a known point x_w (a threshold or a policy change), while the response stays continuous. With $x_2 = \mathbf{1}\{x_1 > x_w\}$,

$$y_i = \beta_0 + \beta_1x_{i1} + \beta_2(x_{i1} - x_w)x_{i2} + \epsilon_i,$$

so that

$$E(y) = \begin{cases} \beta_0 + \beta_1x_1, & x_1 \le x_w, \\ (\beta_0 - \beta_2x_w) + (\beta_1 + \beta_2)x_1, & x_1 > x_w. \end{cases}$$

The two lines meet at x_w, β_2 is the change in slope, and $H_0 : \beta_2 = 0$ tests for a break. The regressor $(x_1 - x_w)_+ = (x_1 - x_w)x_2$ is a **truncated power basis** function; several knots $x_{w_1} < x_{w_2} < \cdots$ add one such term each, giving a linear spline. If a jump at x_w is also allowed, add x_2 itself.

Example 13.2

Unit cost y against lot size x_1 for eight lots, with a suspected change at $x_w = 500$:

x_1	300	340	400	480	570	650	720	800
y	475	440	452	377	355	257	249	139

The fit is $\hat{y} = 589.5 - 0.395x_1 - 0.389(x_1 - 500)_+$: the cost falls by 0.40 per unit up to 500 and by $0.40 + 0.39 = 0.78$ per unit beyond.

13.4 Selected exercises

Motivation. Indicator models are ordinary regressions, so the general results of Chapters 6–8 identify their estimates: a jump test for a broken line, the two-sample t test, the pooled within-group slope, and the equivalence of the interaction model with two separate fits.

Exercise 13.1 (Testing for a jump)

Consider

$$y_i = \beta_0 + \beta_1x_{i1} + \beta_2(x_{i1} - 40)x_{i2} + \beta_3x_{i3} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

where x_1 is quantitative and $x_{i2} = x_{i3} = 1$ if $x_{i1} > 40$ and 0 otherwise. The mean response varies over the range of x_1. Write down hypotheses for testing whether the mean response is discontinuous at $x_1 = 40$, and explain how to carry out the test.

Solution.

Step 1. (The mean function.) For $x_1 \le 40$, $E(y) = \beta_0 + \beta_1 x_1$; for $x_1 > 40$,

$$E(y) = (\beta_0 - 40\beta_2 + \beta_3) + (\beta_1 + \beta_2)x_1.$$

So the model consists of two lines with arbitrary intercepts and slopes on either side of 40; β_2 is the change in slope.

Step 2. (The jump.) The left line at $x_1 = 40$ gives $\beta_0 + 40\beta_1$, and the right line tends to $\beta_0 - 40\beta_2 + \beta_3 + 40(\beta_1 + \beta_2) = \beta_0 + 40\beta_1 + \beta_3$ as $x_1 \downarrow 40$. The jump is β_3, so the hypotheses are

$$H_0 : \beta_3 = 0 \ \text{(continuous at 40)} \qquad \text{vs} \qquad H_1 : \beta_3 \neq 0.$$

Step 3. (The test.) Assume the design matrix with columns $\mathbf{1}$, $\mathbf{x}_1$, $(\mathbf{x}_1 - 40)\mathbf{x}_2$, $\mathbf{x}_3$ has full rank, which holds for instance when there are at least two distinct values of x_1 on each side of 40 (the model is a reparametrization of two separate lines). Fit the model by least squares and compute

$$t_0 = \frac{\hat\beta_3}{\sqrt{c_{33} MSE}}, \qquad MSE = \frac{SSE}{n-4},$$

with c_{33} the diagonal entry of $(X'X)^{-1}$ for β_3. Under H_0, $t_0 \sim t(n-4)$ (Section 6.5); reject when $|t_0| > t_{\alpha/2}(n-4)$. Equivalently (Section 8.2), compare the full model with the continuous broken line of Section 13.3 by the partial F statistic $F_0 = t_0^2$ with $(1, n-4)$ degrees of freedom. ■

Exercise 13.2 (An indicator alone is the two-sample t test)

Let $y_i = \beta_0 + \beta_1 d_i + \epsilon_i$ with $d_i \in \{0, 1\}$, $n_0 \ge 1$ observations with $d = 0$, $n_1 \ge 1$ with $d = 1$, $n = n_0 + n_1 \ge 3$, and group means $\bar y_0, \bar y_1$. Show that $\hat\beta_0 = \bar y_0$, $\hat\beta_1 = \bar y_1 - \bar y_0$, that MSE is the pooled variance s_p^2, and that the t statistic for $\beta_1 = 0$ is the pooled two-sample t statistic.

Solution [Supplement].

Step 1. This is a simple regression on $x = d$ with the two design points $a = 0$, $b = 1$. By Exercise 3.2, $\hat\beta_1 = \bar y_1 - \bar y_0$ and the fitted line passes through $(0, \bar y_0)$, so $\hat\beta_0 = \bar y_0$; also $S_{xx} = n_0 n_1 / n$.

Step 2. The fitted value of each observation is its group mean, so

$$SSE = \sum_{d_i = 0} (y_i - \bar y_0)^2 + \sum_{d_i = 1} (y_i - \bar y_1)^2, \qquad MSE = \frac{SSE}{n-2} = s_p^2.$$

Step 3. By Section 4.1,

$$t_0 = \frac{\hat\beta_1}{\sqrt{MSE / S_{xx}}} = \frac{\bar y_1 - \bar y_0}{s_p \sqrt{n/(n_0 n_1)}} = \frac{\bar y_1 - \bar y_0}{s_p \sqrt{1/n_0 + 1/n_1}},$$

which is the pooled two-sample t statistic, with the same $n - 2$ degrees of freedom. ■

Exercise 13.3 (The common slope of parallel lines)

In the parallel-lines model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$ of Section 13.1, let group $g \in \{0, 1\}$ (with $x_2 = g$) have mean $\bar x_g$ of x_1, mean $\bar y_g$, and within-group sums $S_{xx,g} = \sum_{i \in g} (x_{i1} - \bar x_g)^2$, $S_{xy,g} = \sum_{i \in g} (x_{i1} - \bar x_g)(y_i - \bar y_g)$, with $S_{xx,0} + S_{xx,1} > 0$. Show that

$$\hat\beta_1 = \frac{S_{xy,0} + S_{xy,1}}{S_{xx,0} + S_{xx,1}}, \qquad \hat\beta_2 = (\bar y_1 - \bar y_0) - \hat\beta_1 (\bar x_1 - \bar x_0).$$

Solution [Supplement].

Step 1. Apply Theorem 6.5 with $X_1 = (\mathbf{1}, \mathbf{x}_2)$ and $X_2 = \mathbf{x}_1$. The column space of X_1 is spanned by the group indicators $\mathbf{1}_{G_0} = \mathbf{1} - \mathbf{x}_2$ and $\mathbf{1}_{G_1} = \mathbf{x}_2$, which are orthogonal. So $H_1\mathbf{v}$ replaces each entry of $\mathbf{v}$ by the mean of its group, and $(I - H_1)\mathbf{v}$ subtracts the group means.

Step 2. Hence $\mathbf{e} = (I - H_1)\mathbf{y}$ has entries $y_i - \bar{y}_{g(i)}$ and $X_{2\cdot1} = (I - H_1)\mathbf{x}_1$ has entries $x_{i1} - \bar{x}_{g(i)}$. Since $\|X_{2\cdot1}\|^2 = S_{xx,0} + S_{xx,1} > 0$, the design has full rank, and

$$\hat{\beta}_1 = \frac{X_{2\cdot1}'\mathbf{e}}{X_{2\cdot1}'X_{2\cdot1}} = \frac{S_{xy,0} + S_{xy,1}}{S_{xx,0} + S_{xx,1}}.$$

Step 3. The residuals of the full fit are orthogonal to $\mathbf{1}_{G_0}$ and $\mathbf{1}_{G_1}$, so they sum to zero within each group. Averaging the fitted equation over each group gives $\bar{y}_0 = \hat{\beta}_0 + \hat{\beta}_1\bar{x}_0$ and $\bar{y}_1 = \hat{\beta}_0 + \hat{\beta}_2 + \hat{\beta}_1\bar{x}_1$. Subtracting yields $\hat{\beta}_2$.

So the common slope pools the within-group information only, and $\hat{\beta}_2$ compares the group means *adjusted* to a common value of x_1 (the analysis of covariance). ■

Exercise 13.4 (The interaction model is two separate fits)

Fit the interaction model $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ ($x_2 \in \{0, 1\}$) to the pooled data, and separately fit lines $b_{0g} + b_{1g}x_1$ in each group g (each group having at least two distinct x_1 values). Show that both give the same fitted values and $SSE = SSE_0 + SSE_1$, and that $\hat{\beta}_0 = b_{00}$, $\hat{\beta}_1 = b_{10}$, $\hat{\beta}_2 = b_{01} - b_{00}$, $\hat{\beta}_3 = b_{11} - b_{10}$.

Solution [Supplement].

Step 1. Let Z have columns $\mathbf{1}_{G_0}$, $\mathbf{x}_1\mathbf{1}_{G_0}$, $\mathbf{1}_{G_1}$, $\mathbf{x}_1\mathbf{1}_{G_1}$ (products taken entrywise), the design of the separate fits. The interaction design has columns

$$\mathbf{1} = \mathbf{1}_{G_0} + \mathbf{1}_{G_1}, \quad \mathbf{x}_1 = \mathbf{x}_1\mathbf{1}_{G_0} + \mathbf{x}_1\mathbf{1}_{G_1}, \quad \mathbf{x}_2 = \mathbf{1}_{G_1}, \quad \mathbf{x}_1\mathbf{x}_2 = \mathbf{x}_1\mathbf{1}_{G_1},$$

so $X = ZA$ for a 4×4 matrix A; it is nonsingular, since conversely $\mathbf{1}_{G_0} = \mathbf{1} - \mathbf{x}_2$ and $\mathbf{x}_1\mathbf{1}_{G_0} = \mathbf{x}_1 - \mathbf{x}_1\mathbf{x}_2$ express Z through X.

Step 2. Ordering the observations by group, Z is block diagonal, so $\|\mathbf{y} - Z\mathbf{b}\|^2 = \sum_g \|\mathbf{y}_g - b_{0g}\mathbf{1} - b_{1g}\mathbf{x}_{1,g}\|^2$ is minimized by minimizing each term separately: the least squares fit on Z is the pair of separate lines, with $SSE = SSE_0 + SSE_1$. Each block has full rank by the assumption on the x_1 values, so Z and hence X have full rank.

Step 3. By Exercise 6.3, X and Z give the same fitted values and SSE, so $X\hat{\beta} = Z\hat{\mathbf{b}}$. By Step 1,

$$X\hat{\beta} = \hat{\beta}_0\mathbf{1}_{G_0} + \hat{\beta}_1\mathbf{x}_1\mathbf{1}_{G_0} + (\hat{\beta}_0 + \hat{\beta}_2)\mathbf{1}_{G_1} + (\hat{\beta}_1 + \hat{\beta}_3)\mathbf{x}_1\mathbf{1}_{G_1},$$

and since the columns of Z are linearly independent, $b_{00} = \hat{\beta}_0$, $b_{10} = \hat{\beta}_1$, $b_{01} = \hat{\beta}_0 + \hat{\beta}_2$ and $b_{11} = \hat{\beta}_1 + \hat{\beta}_3$. This justifies the remark after the interaction model in Section 13.1. ■